



V₂CT_x MXene — Multilayer Powder

Research-grade two-dimensional transition-metal carbide powder

V₂CT_x is a two-dimensional transition-metal carbide MXene produced from a V₂AlC precursor. This multilayer powder is intended for research requiring a conductive, layered carbide material with micron-scale dimensions. Surface terminations include –O, –OH, –F, and –Cl; their relative distribution varies with synthesis and lot.

KEY FEATURES

- Layered V₂CT_x morphology derived from V₂AlC
- Micron-scale nominal flake / particle size
- Typical conductivity of 10–100 S/cm
- Mixed –O/–OH/–F/–Cl surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal–sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	V ₂ CT _x MXene
Precursor	V ₂ AlC
Product form	Multilayer powder
Preparation	HF etching
Layer structure / count	Multilayer
Nominal flake / particle size	2–20 μm
Typical conductivity	10–100 S/cm
Surface terminations	–O, –OH, –F, and –Cl
Appearance	Dark brown / brown-black powder (descriptive only)
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION

	Multilayer V ₂ CT _x — SEM. Representative characterization; not lot-specific.
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Product positioning: Supplied and technically supported by WiniGen Materials for research and development use.

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V2CTX-ML-TDS-001

Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.